Induction Heads and Model Collapse: How the Real-to-Synthetic Data Ratio Shapes In-Context Learning Circuits

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Abstract

Model collapse has so far been measured through outputs, which may understate it if internal mechanisms degrade first. This study therefore traces the induction circuit behind in-context learning (ICL) in two-layer transformers retrained on data that the previous model partly or wholly relabels, or that accumulates alongside the real data. A tuned model reproduces its own labels almost exactly, so that accuracy holds under every real-to-synthetic ratio. Once errors are forced through a raised sampling temperature, however, mechanistic changes precede accuracy loss: the logit margin halves and the circuit spreads while accuracy is almost unchanged, and the margin predicts the next generation's error rate, so that errors compound. Retaining real data limits how far accuracy falls, and accumulation delays its decline, but neither stops the margin's erosion. Overall, retaining real data protects a model's behaviour for longer than its circuit, so that collapse may be detected in the circuit before accuracy falls.

Keywords: model collapse; induction heads; mechanistic interpretability

Introduction

Recursively training large language models (LLMs) on data generated by other models causes *model collapse*, a loss of information about the true data distribution (Shumailov et al., 2024), which retaining real data appears to avoid (Gerstgrasser et al., 2024). Both findings, however, rest on outputs, which would understate collapse if internal mechanisms degrade first.

In-context learning (ICL) lets models adapt to new tasks from prompt examples without parameter updates (Olsson et al., 2022; Singh et al., 2024), and in the task of Singh et al. (2024) it is carried out by an induction circuit whose components are known, so that damage can be traced to specific heads rather than inferred from accuracy. This study therefore asks how the real share of each training pool shapes this circuit.

Background

Induction heads A previous-token head copies each symbol into the label that follows it, and an induction head matches the query against these copies and copies the matching label; several such heads can contribute additively (Singh et al., 2024), and their formation tracks ICL ability (Chen et al., 2026).

Collapse and accumulation Collapse arises mainly from sampling error, which loses the tails of the distribution first and compounds across generations (Shumailov et al., 2024). Replacing real data makes test error grow with every iteration, whereas accumulating it keeps the error bounded (Gerstgrasser et al., 2024), a contrast tested here in the circuit.

Methods

This study uses the setup of Singh et al. (2024): each sequence holds two exemplar–label pairs and a query, with Omniglot exemplars (Lake et al., 2015) encoded by a frozen ResNet18 (He et al., 2016) and five labels reassigned per sequence, so that only the context holds the answer. A causal two-layer attention-only transformer with eight heads per layer trains on 50 classes for one million sequences, at learning rates of 10^{-4} (base) and 3×10^{-4} (extended) tuned on 1,473 validation classes, and is tested on 100 unseen classes. The base model predicts the query's label, while the extended model also predicts a next symbol, taken from the context in real data but sampled by the model over all 50 training classes for synthetic data, and its label, so that three tokens per sequence become synthetic.

Real-to-synthetic ratio Each run trains on a pool of $N = 20{,}000$ sequences for up to 20 generations, the first real, with training length fixed and three seeds per condition. The previous model then regenerates the targets of 0%, 50%, 90%, or 100% of the pool from their real prompts, where 90% matches the 10% of real data kept by Shumailov et al. (2024). Accumulation instead adds N synthetic sequences per generation (Gerstgrasser et al., 2024), matching 50% replacement at generation 1, so that keeping earlier data can be separated from the real share. As the tuned model makes almost no errors when sampling targets at $T = 1$, errors are also forced at $T = 3$ and 5.

Circuit analysis Each generation is analysed on the same real sequences through its margin, the logit lead of the correct label over the other label in the context, and the edges that Automatic Circuit DisCovery (ACDC) (Conmy et al., 2023) keeps at $\tau = 0.001$ against copies in which the two context exemplars swap places, which measure how strongly and how sparsely the circuit separates the answer. For the induction head, its attention to the matching label measures the match, and its direct effect on the margin the copy.

Results

Table 1: Generation 19 of every 20-generation run, mean ± SD over three seeds, under full replacement unless stated. †Margin below 1, where the circuit size is less reliable.

		Base model			Extended	
Condition	Real	Acc.	Margin	Edges	Margin	Edges
No forced errors ($T = 1$)						
Real only	100%	0.98 ±0.01	17.5 ±2.2	23 ±5	15.7 ±0.3	42 ±7
100% replaced	0%	0.98 ±0.00	12.5 ±0.3	35 ±7	12.8 ±0.9	92 ±36
Accumulate	5%	0.99 ±0.00	9.8 ±1.0	20 ±3	6.3 ±0.1	34 ±5
Forced errors ($T = 3$)						
50% replaced	50%	0.79 ±0.01	2.6 ±0.1	194 ±11	3.8 ±0.3	89 ±19
90% replaced	10%	0.33 ±0.01	0.4 ±0.1	207 ±3†	0.1 ±0.0	180 ±4†
100% replaced	0%	0.19 ±0.02	0.1 ±0.0	207 ±7†	−0.1 ±0.1	163 ±7†
Accumulate	5%	0.92 ±0.02	0.7 ±0.0	17 ±5†	0.6 ±0.0	11 ±2†

No errors, no collapse At generation 0, both models solve the task by ICL: the base model reaches a mean test accuracy of 0.98 and the extended model 0.96, with validation accuracies of 0.99 and 0.97 respectively. Without forced errors, the ratio barely matters, as at $T = 1$ only 0.01% of regenerated labels are wrong after 20 generations, so that every replacement level yields almost the same model (Table 1).

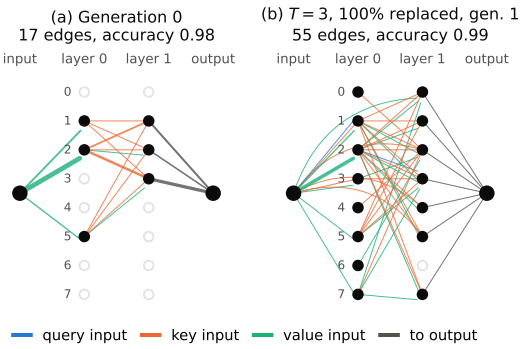


Figure 1: ACDC circuits of the base model (one seed), with filled nodes for heads in the circuit: the previous-token head A0.2 feeds the induction head A1.3 (a), and one generation at $T = 3$ triples it (b).

Confidence and circuit change before accuracy Once errors are forced at $T = 3$, the model's confidence and its circuit change well before it starts answering wrongly. After one generation of full replacement, its margin has roughly halved, from 18.4 to 10.1, and its circuit has more than tripled, from 22 to 74 edges on average (17 and 55 in Figure 1), yet accuracy is still 0.95 or above. Only by generation 3, with the circuit spread across 211 edges, does accuracy fall, to 0.66. This growth is not an artefact of ACDC's fixed threshold: a threshold-free count, the edges needed to recover 80% of the margin, rises from 33 to 120, against 30 under real-only retraining.

The match holds, the copy weakens Inside the induction head, its two steps fail at different times (Figure 2). Under full replacement, the head still attends to the matching label, keeping 88% of its generation 0 attention by generation 2, whereas what it writes to the output, its direct effect, keeps only 41%, so that it still finds the answer but copies it far more weakly. Its attention collapses only at generation 4, after accuracy. Under accumulation, attention stays at 0.89 or above for 15 generations while the direct effect falls by 86%, so that real data preserve where the head looks, not how strongly it copies.

The margin sets the next generation's errors Each model labels the next generation's data, so that its confidence decides how many of those labels are wrong. Taking its mean margin D_g over the other context label as its lead over all four wrong labels, sampling at T makes a share

$$\hat{\varepsilon}_{g+1} = \frac{4\,e^{-D_g/T}}{1+4\,e^{-D_g/T}} \tag{1}$$

of these labels wrong. This approximation predicts the share of wrong labels in each new pool to within 2.1 percentage points on average across both models, temperatures, and conditions ($r = 0.994$). A falling margin therefore writes more errors into the next pool, which lowers the next margin in turn, a loop that only the correct labels of real data can slow.

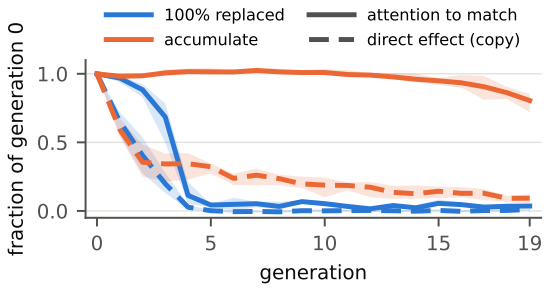


Figure 2: Induction-head match and direct effect at $T = 3$, relative to generation 0, over three seeds.

The real share sets how far collapse goes Over 20 generations at $T = 3$, the more real data a pool keeps, the higher accuracy settles: 0.79 when half the pool stays real, 0.33 with a tenth, and 0.19 with none (Table 1). Yet even half leaves the circuit spread across 194 edges, so real data bound accuracy without keeping the circuit intact. Accumulation holds accuracy at 0.96 or above for 15 generations while its margin falls by 96%, and accuracy declines only once the margin drops below 1. The extended model, whose query accuracy settles at 0.84, 0.34, and 0.20, shows the same split in a second output: under accumulation, its query accuracy stays at 0.93, but its choice of next symbol, scattered by the raised temperature from generation 1, keeps only 13% of its probability on the context symbols at $T = 1$, against 98.6% under real-only retraining.

Discussion

The real-to-synthetic ratio shaped the induction circuit mainly once the synthetic data carried errors, and then in two ways. The share of real data set how far accuracy and the margin fell, as only real labels dilute the errors that a falling margin writes into each new pool, which extends Shumailov et al. (2024) with the internal quantity that sets the sampling error. At every ratio, however, the circuit changed before accuracy did: the margin fell, the circuit spread, and within the induction head the copy weakened while the match held, suggesting that the copying operation of Singh et al. (2024) is the more fragile. Accumulation protected accuracy but not the margin or the copy, consistent with the bounded error of Gerstgrasser et al. (2024). The margin and the circuit could therefore flag collapse before accuracy does. These conclusions are bounded by forcing errors through the sampling temperature rather than through a weaker model, and by a training length fixed while the accumulated pool grows 20-fold.

Conclusion

This investigation showed that, once synthetic data carry errors, confidence and the induction circuit change before accuracy, while the share of real data sets only how high accuracy settles. Overall, real data protect what a model does for longer than how confidently it does it, so that collapse is better detected in the circuit than in accuracy.

Contribution Statements

We, Lily de Melo, Tawananyasha Kadango, and Gopolang Mohlakola, worked together fairly on this project.

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    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: [NA]

    Justification: This study involves neither crowdsourcing nor research with human subjects.

    Guidelines:

    - The answer NA means that the paper does not involve crowdsourcing nor research with human subjects.
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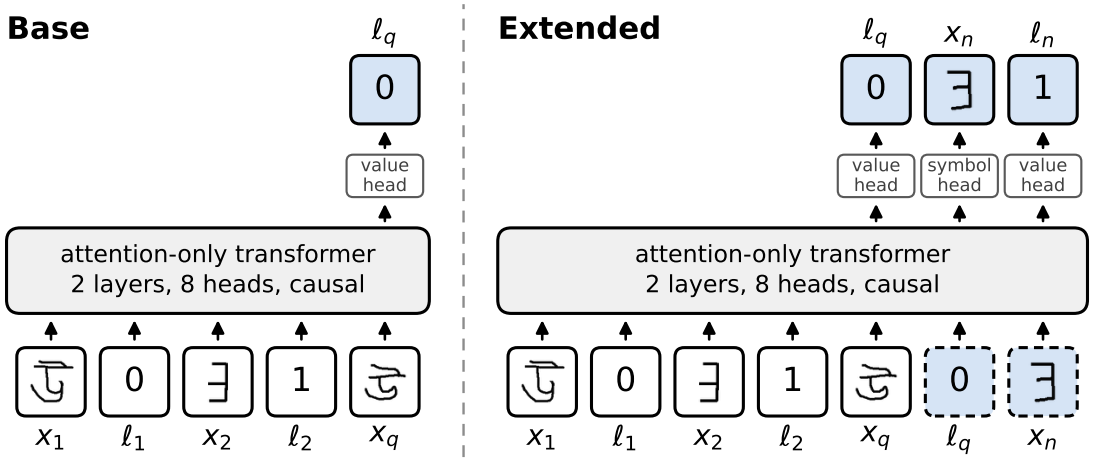


Figure S1: Base (left) and extended (right) models. The base model predicts the query's label from two symbol–label pairs and the query, and the extended model also predicts the next symbol and its label. Shaded boxes are model outputs; dashed inputs are true tokens during training and the model's own samples when it generates the next generation's data.

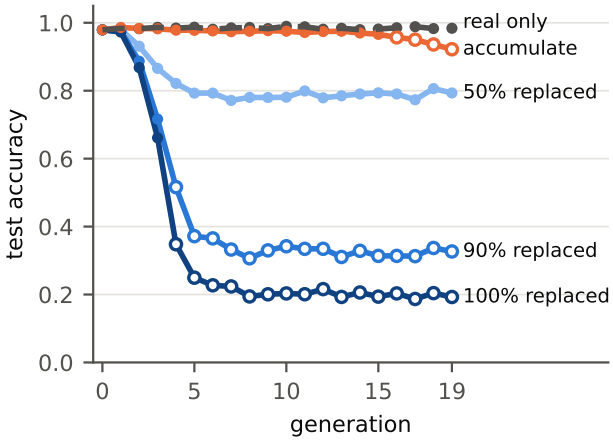


Figure S2: Test accuracy of the base model at $T = 3$, mean of three seeds. Each replacement level settles at an accuracy set by its real share, while accumulation stays with the real-only control for about 15 generations before declining. Hollow markers show a mean margin below 1.

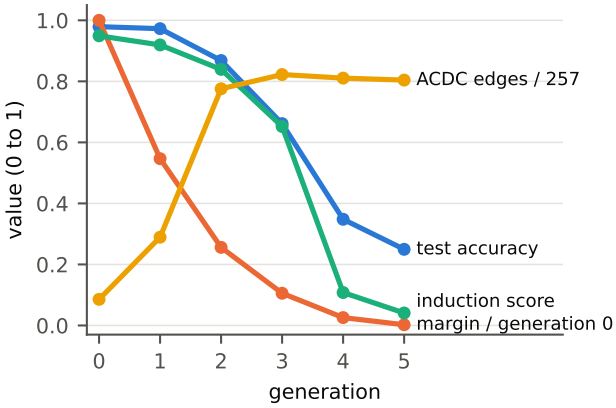


Figure S3: Order of failure of the base model under full replacement at $T = 3$, mean of three seeds. The margin halves and the circuit more than triples by generation 1, whereas accuracy and the induction score fall mainly from generation 3.

Wits University Faculty of Science post-graduate student AI declaration

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I declare that in the course of conducting the research towards my degree or in the preparation of this thesis/dissertation/research report (select one by marking with an X):

I **did not** make use of generative AI tools [ ]

I **did** make use of generative AI tools for the following (tick all that apply):

1.	Idea Generation (research problem/design, hypothesis)	[ ]
2.	Sourcing Related Work (summarising, identifying sources)	[ ]
3.	Methods and Experiment Design (experiment setup, model tuning)	[X]
4.	Data Analysis (presentation, coding, interpretation)	[X]
5.	Theoretical Development (theorem proving, conceptual analysis)	[ ]
6.	Code Development (generating algorithms, writing scripts)	[X]
7.	Presentation (rendering graphics, formatting)	[X]
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